Agents as Personal Assistants: A Research Survey for the University of Passau Seminar

TL;DR

- **The field has pivoted from intent-classification chatbots (Siri/Alexa era) to tool-using LLM agents** that interleave reasoning, planning, and tool calls (ReAct, Yao et al. ICLR 2023), with email and calendar serving as the canonical "personal-assistant" tool surfaces; the conceptual core problem has shifted from natural-language understanding to *trustworthy autonomous action over private data*.
- **Privacy is now the dominant research bottleneck**, not capability: indirect prompt injection (Greshake et al. AISec 2023), contextual-integrity violations (Mireshghallah et al. ICLR 2024; Shao et al. NeurIPS 2024), and the "confused deputy" problem mean frontier models leak sensitive information in 25–57% of agentic tasks even with privacy-enhancing prompts.
- **Benchmarks remain the field's weakest link**: WorkBench, τ-bench, AgentDojo, GAIA, PrivacyLens, and ConfAIde each isolate one facet (utility, robustness, privacy), but no single benchmark jointly evaluates long-horizon email/calendar workflows under realistic adversarial and personalization pressure — this is the open research frontier the seminar should foreground.

Key Findings

1. The "agent" abstraction in personal-assistant research is now defined by five capability axes: *state-reading, planning, tool-use, memory, and human handoff* — formalized in the Personal LLM Agents survey (Li et al. 2024, arXiv:2401.05459).
2. The dominant architecture is the ReAct loop (Yao et al. ICLR 2023); more advanced systems layer planner–executor, supervisor–subagent, and reflection components on top.
3. Email research has a long history (Smart Reply, Kannan et al. KDD 2016; Smart Compose, Chen et al. KDD 2019) that pre-dates LLM agents but informs current work on tone matching, action-item extraction, and draft generation.
4. Calendar research has an even longer pedigree (CALO/PTIME, Berry et al. ACM TIST 2011) emphasizing preference learning under constraint satisfaction — concerns largely *absent* from modern LLM scheduling work, which is a rediscovery opportunity.
5. Memory is the central personalization mechanism. Generative Agents (Park et al. UIST 2023) introduced the memory-stream + reflection + planning triad; MemGPT/Letta (Packer et al. 2023) operationalized OS-style hierarchical memory; Mem0 (Chhikara et al. ECAI 2025) and A-MEM (2025) push toward production-grade memory with graph structure.
6. Industry's privacy answer is *architectural*: Apple's Private Cloud Compute (2024) and

on-device Apple Intelligence operationalize "stateless computation on personal user data" with verifiable attestation — currently the most ambitious deployed personal-assistant privacy stack.

7. The field lacks consensus on evaluation: deterministic state-checking benchmarks (WorkBench, AgentDojo) and LLM-judged simulations (ToolEmu, τ-bench) disagree on what "success" means.

Details

1. Conceptual Foundations and Taxonomy

Defining the personal-assistant agent. Marketing usage equates "personal assistant" with any conversational interface (Siri, Alexa, ChatGPT). The research literature is sharper. Li et al. (*Personal LLM Agents: Insights and Survey about the Capability, Efficiency and Security*, arXiv:2401.05459, 2024) define a Personal LLM Agent as "LLM-based agents that are deeply integrated with personal data and personal devices and used for personal assistance." This is distinct from a generic chatbot because it (a) reads private state (email, calendar, contacts, location), (b) plans multi-step actions, (c) invokes external tools/APIs, (d) maintains memory across sessions, and (e) escalates to the user under uncertainty. The survey synthesizes opinions from 25 domain experts and organizes the field around three axes: capability, efficiency (latency/footprint), and security/privacy. arXiv

The 2022 shift. Pre-2022, personal-assistant pipelines were *intent classifiers* — neural classifiers mapping user utterances to slot-filled intents (the Alexa/Siri DialogFlow paradigm). Post-2022, the field pivoted to *tool-using LLMs* prompted to reason in natural language and call APIs as needed. The pivot paper is **Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models"** (ICLR 2023, arXiv:2210.03629), which proposed interleaving "Thought → Action → Observation" steps in the prompt. ReAct outperformed prior baselines on HotpotQA, Fever, ALFWorld, and WebShop by absolute success-rate margins of 10–34%, and has become the de facto default loop in LangChain/LangGraph agents. arXiv arXiv

Core architectural patterns seen across the literature:

- *ReAct loop* (Yao et al. 2023): single agent, thought-action-observation, no separate planner.
- *Planner–Executor*: a slow "planner" LLM emits a plan; a faster "executor" runs steps and can replan.
- *Supervisor–Subagent / Multi-agent*: a router LLM dispatches to specialized agents (e.g., ScheduleMe, Wijerathne et al. PACLIC 2025, arXiv:2509.25693, which uses GPT-4o mini as supervisor over calendar specialists).
- *Memory-augmented*: MemGPT-style hierarchical memory, retrieval-augmented context (Packer et al. arXiv:2310.08560, 2023).
- *Reflection / self-critique*: Generative Agents (Park et al. UIST 2023) reflect periodically

over the memory stream.

Required capabilities for a personal assistant (synthesizing Li et al. 2024 and the WorkBench task taxonomy):

1. *State-reading* (calendar/email APIs, mailbox search).
2. *Planning* (multi-step task decomposition).
3. *Tool use* (function calling, MCP servers).
4. *Memory* (preferences, recurring contacts, prior decisions).
5. *Human handoff* (uncertainty → confirmation).

2. Email Agents — Research Focus

The pre-LLM canon. Two Google papers anchor the field:

- **Kannan et al., "Smart Reply: Automated Response Suggestion for Email"** (KDD 2016, arXiv:1606.04870). A sequence-to-sequence LSTM trained on Gmail proposes three short replies. As reported by the authors: "The system is currently used in Inbox by Gmail and is responsible for assisting with 10% of all mobile responses. It is designed to work at very high throughput and process hundreds of millions of messages daily." Introduced engineering challenges still relevant today: response diversity (avoid three near-identical suggestions), semantic clustering of replies, and large-scale inference. arxiv arXiv
- **Chen et al., "Gmail Smart Compose: Real-Time Assisted Writing"** (KDD 2019, arXiv:1906.00080). A hybrid bag-of-words-over-context + LSTM-over-prefix architecture for inline auto-completion at sub-100ms latency. Crucially, this paper foregrounded *personalization via per-user language-model adaptation* and *fairness/bias filtering* — both still open in 2026. ACM SIGKDD

LLM-era email agents (2023–2026). The literature is thinner than one might expect:

- **Styles et al., "WorkBench: a Benchmark Dataset for Agents in a Realistic Workplace Setting"** (arXiv:2405.00823, 2024) defines 690 tasks across email, calendar, CRM, analytics, and project-management tools with outcome-centric (deterministic database-state) evaluation. The headline result is sobering: GPT-4 succeeds on only 43% of tasks; Llama-2-70B on 3%; and errors frequently take the *wrong* action (e.g., emailing the wrong recipient). OpenReview arXiv
- **Wu et al., "Control at Stake: Evaluating the Security Landscape of LLM-Driven Email Agents"** (arXiv:2507.02699, 2025; preprint) systematically surveys deployed LLM email agents and shows that most can be hijacked by indirect prompt injection embedded in incoming mail — exfiltrating data or sending unauthorized messages. arxiv
- **Shao et al., PrivacyLens** (NeurIPS 2024 Datasets & Benchmarks, arXiv:2409.00138) shows GPT-4 agents leak sensitive information in ~25% of email-drafting tasks even when explicitly prompted to be privacy-aware.

Open challenges in email agents:

- *Hallucination in drafting* (fabricating facts about prior interactions);
- *Tone matching* across recipients (formal/informal, native-language register);
- *Recipient privacy* (the agent must not quote third-party emails in replies to others);
- *Thread context handling* (long threads exceed context windows; summarization loses signal);
- *Multi-message reasoning* (cross-referencing past commitments).

3. Calendar / Scheduling Agents — Research Focus

The classic CALO line. SRI's DARPA-funded **Cognitive Assistant that Learns and Organizes** (CALO) is the most ambitious personal-assistant research program ever attempted, and its scheduling subsystem is uniquely well-documented. As SRI's official program page states, CALO was "a five-year, $150 million project that brought together more than 300 researchers from 22 premier research institutions." Key scheduling publications: SRI International

- **Berry, Gervasio, Peintner, Yorke-Smith, "PTIME: Personalized Assistance for Calendaring"** (ACM TIST 2(4), Article 40, 2011). PTIME combines a constraint-based scheduler with the **PLIANT** online preference learner (SVM-based) that unobtrusively learns scheduling preferences from user choices. It interoperates with Outlook/Exchange and handles meeting requests over email. Tudelft
- **Gervasio et al., "Active Preference Learning for Personalized Calendar Scheduling Assistance"** (IUI 2005) formalizes the preference-elicitation problem.
- The **CALO-MA** meeting-assistant component focused on multi-party meetings with speech and language processing.
- **PExA** (Peintner et al., IUI 2008 demo) is the SPARK-BDI-based personal desktop assistant that wrapped the CALO components for end-users.

The x.ai post-mortem. x.ai (Mortensen and Jimenez Belenguer, founded 2014) offered "Amy" and "Andrew" — email-based scheduling assistants the user CC'd into a thread. Founder Dennis Mortensen confirmed in his Growth Everywhere podcast interview: "They had to raise $44,000,000 because humans are dynamic and language is difficult to parse." Despite that funding and ~100 human labelers, the product was shut down on October 31, 2021 following acquisition by Bizzabo. There is no peer-reviewed paper, but the shutdown is a primary teaching artifact: it illustrates that open-domain NLU for scheduling was *too brittle* even at massive labeled-data scale — a problem LLMs partly but not fully solve. Leveling Up TASBIA

Modern LLM scheduling agents.

- **ScheduleMe** (Wijerathne et al., PACLIC 2025, arXiv:2509.25693): LangGraph supervisor agent (GPT-4o mini, ReAct) routing to calendar specialists; evaluated zero-shot on 120 cases. WSO2

- AgentDojo's *workspace* environment (Debenedetti et al. NeurIPS 2024) and WorkBench (Styles et al. 2024) both include calendar tools; agents fail unsafely under adversarial conditions.

Open challenges in scheduling agents:

- *Time-zone reasoning* (LLMs frequently mishandle DST transitions and IANA zones);
- *Preference elicitation* (cold-start, implicit feedback);
- *Multi-party coordination* (negotiation across agents with conflicting calendars);
- *Constraint satisfaction* (LLMs cannot reliably solve even simple CSPs without external solvers);
- *Temporal reasoning generally* — TimeBench (Chu et al. arXiv:2311.17667), EvolveBench (ACL 2025), and TimeShift (arXiv:2409.13338) all show frontier LLMs are brittle on time-sensitive recall and reasoning.

4. Privacy Considerations — Research Focus

Why personal assistants are uniquely privacy-critical. They simultaneously hold (a) the user's complete written communication history, (b) their physical and temporal whereabouts (calendar, location), (c) their contact graph, and (d) the authority to *act* on those data — a combination no prior software category has had. Mireshghallah et al. (ConfAIde, ICLR 2024) put it plainly: "LLMs are fed different types of information from multiple sources … and are expected to reason about what to share in their outputs, for what purpose and with whom, within a given context."

Indirect prompt injection — the foundational attack. Greshake et al., "Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection" (Proc. 16th ACM Workshop on Artificial Intelligence and Security / AISec '23; arXiv:2302.12173) showed that an attacker can plant instructions in *data* the LLM later retrieves (a webpage, a calendar event, an incoming email) and the LLM will follow them. For a personal assistant that reads attacker-controlled email, this is catastrophic: any third party can become a co-pilot. The paper derives a security taxonomy covering data theft, worming, ecosystem contamination, and fraud. arXiv arXiv

Benchmarks for prompt injection in agents:

- **Debenedetti et al., "AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents"** (NeurIPS 2024 Datasets & Benchmarks Track, arXiv:2406.13352). 70 tools, 97 user tasks, 629 security cases across workspace, banking, Slack, and travel environments. The paper reports: "our attacks succeed against the best performing agents in less than 25% of cases. When deploying existing defenses against prompt injections, such as a secondary attack detector, the attack success rate drops to 8%." AgentDojo's key methodological contribution is

deterministic utility/security evaluation over environment state, rather than LLM-judged simulation (which itself can be injected). arXiv arxiv

- **Zhan et al., "InjecAgent"** (ACL Findings 2024, arXiv:2403.02691). 1,054 test cases across 17 user tools and 62 attacker tools; ReAct-prompted GPT-4 is vulnerable to indirect prompt injection ~24% of the time, and the authors report that "the incorporation of the 'hacking prompt' further increases its success rate to 47%." arXiv + 3
- **Ruan et al., "Identifying the Risks of LM Agents with an LM-Emulated Sandbox"** (ToolEmu, ICLR 2024 Spotlight, arXiv:2309.15817). 36 high-stakes tools, 144 cases; even the safest agent fails 23.9% of the time.

Contextual Integrity as the normative framework. Helen Nissenbaum's *Contextual Integrity* (CI) theory — privacy as *appropriate information flow* governed by context-specific norms (sender, recipient, subject, attribute, transmission principle) — has become the dominant theoretical lens for agent privacy:

- **Mireshghallah et al., "Can LLMs Keep a Secret? Testing Privacy Implications of Language Models via Contextual Integrity Theory"** (ConfAIde, ICLR 2024 Spotlight, arXiv:2310.17884). A 4-tier benchmark where GPT-4 leaks private info inappropriately 39% of the time, ChatGPT 57% — even with privacy-inducing prompts.
- **Shao et al., "PrivacyLens"** (NeurIPS 2024 D&B, arXiv:2409.00138). Extends CI to *agent trajectories*, not just probing questions: GPT-4 agents leak in 25.68% of agentic email/post-composition tasks; Llama-3-70B in 38.69%. Critically, it shows a *gap between answering "should you share X?" correctly and actually not sharing X when executing a task*.
- **Bagdasarian et al., "AirGapAgent: Protecting Privacy-Conscious Conversational Agents"** (arXiv:2405.05175, 2024) operationalizes CI by limiting agent data access to the minimum necessary per task.
- **Ghalebikesabi et al., "Operationalizing Contextual Integrity in Privacy-Conscious Assistants"** (arXiv:2408.02373, 2024).

Industry's privacy architecture: Apple Private Cloud Compute (2024). Apple's PCC is the most technically ambitious deployment of privacy-preserving cloud AI to date. Apple's published security overview lists the core guarantees verbatim: *stateless computation on personal user data* ("Private Cloud Compute must use the personal user data that it receives exclusively for the purpose of fulfilling the user's request"); *user-device cryptographic attestation* of the server cluster before sending requests; *no-administrative-access* server design (no SSH, no remote shells); and *verifiable transparency* via signed binaries available to independent researchers (though not full source code). Apple deliberately couples on-device Apple Intelligence (a ~3-billion-parameter foundation model) for routine tasks with PCC offload for harder ones; only data *relevant* to the request is sent. Apple + 3

The confused-deputy problem. A personal assistant has the user's authority but executes instructions that may originate from third parties (incoming emails). This is the classical confused-deputy pattern from operating-systems security. AgentDojo and InjecAgent both essentially measure how often the deputy is confused. Defenses include *tool filtering*, *task-aligned shielding* (Task Shield, 2024), and *capability scoping*.

5. Personalization — Research Focus

The memory stack. Modern personal-assistant personalization is a memory problem. The canonical paper is **Park, O'Brien, Cai, Morris, Liang, Bernstein, "Generative Agents: Interactive Simulacra of Human Behavior"** (UIST 2023, arXiv:2304.03442). It introduced three components — (a) the *memory stream* (a chronological natural-language record of observations), (b) *reflection* (the LLM periodically synthesizes higher-level insights), and (c) *planning* (which uses memories and reflections to generate the day's plan). Park et al. demonstrated emergent social behavior across 25 Sims-style agents and showed via ablation that each component contributes critically to believability. 3dvar 3dvar

Memory architectures derived from this lineage:

- **MemGPT/Letta** (Packer, Wooders, Lin, Fang, Patil, Stoica, Gonzalez; arXiv:2310.08560, 2023): treats the LLM context window as RAM and external storage as disk; the LLM uses function calls to *page* between core, recall, and archival memory. Now part of the Letta open-source framework. Letta Leonie Monigatti
- **Mem0** (Chhikara, Khant, Aryan, Singh, Yadav; arXiv:2504.19413, accepted at ECAI 2025): a production-oriented memory layer with single-pass extraction, multi-signal retrieval (semantic + BM25 + entity), and a graph-augmented variant Mem0g. Reports ~91% lower p95 latency and ~90% token-cost reduction vs full-context baselines on the LoCoMo benchmark.
- **A-MEM** (Xu et al., arXiv:2502.12110, 2025): agentic memory that dynamically forms its own links between memories rather than relying on predefined schemas.

Preference learning and the cold-start problem. Classic work (Gervasio et al. IUI 2005; PTIME 2011) used SVM-based active preference learning. Modern LLM agents largely *substitute prompt engineering for preference learning* — encoding preferences in the system prompt or core memory block. This is brittle, doesn't generalize, and tends to forget preferences not retrieved into context. The research community has not converged on a replacement for active preference learning in the LLM era.

Style/tone adaptation. Smart Compose's per-user LM adaptation (Chen et al. 2019) remains a strong baseline. Recent LLM work uses few-shot exemplars (the user's prior emails) loaded into context; this leaks the user's writing style into prompts and creates a tension with privacy-preserving inference.

The personalization–privacy tension. Stronger personalization requires more data; more data raises confidentiality risk. Apple's resolution is architectural (on-device + PCC); the research community's resolution is technical (federated learning, differential privacy, on-device LoRA fine-tuning, capability-scoped memory). Brown et al., "What Does It Mean for a

Language Model to Preserve Privacy?" (arXiv:2202.05520, 2022) is the canonical position paper here.

6. Evaluation and Benchmarks

The field is in a benchmark-fragmentation phase. The major artifacts:

Benchmark	Year/Venue	Focus	Eval style	Headline result
τ-bench (Yao et al., Sierra, arXiv:2406.12045)	2024	Customer-service tool agents in airline/retail	LLM-simulated users + deterministic state checks	"Even state-of-the-art function calling agents (like gpt-4o) succeed on <50% of the tasks, and are quite inconsistent (pass^8 <25% in retail)" arxiv
AgentDojo (Debenedetti et al., NeurIPS 2024 D&B)	2024	Prompt-injection robustness	Deterministic state checks	Best agents attacked <25% before defenses; 8% under secondary detector; arxiv tool filtering reduces ASR to 7.5% but utility drops from 69% to 53.3% on GPT-4o
WorkBench (Styles et al., arXiv:2405.00823)	2024	Workplace knowledge-worker tasks (email, calendar, CRM)	Outcome-centric (DB state)	GPT-4: 43%; Llama-2-70B: 3%
ToolEmu (Ruan et al., ICLR 2024 Spotlight)	2024	Safety risks of tool use	LM-emulated sandbox + LM judge	Safest agent fails 23.9%; 68.8% of identified failures are real
InjecAgent (Zhan et al., ACL Findings 2024)	2024	Indirect prompt injection	Trajectory inspection	ReAct GPT-4: 24% vulnerable; 47% with hacking prompt arXiv
ConfAIde (Mireshghallah et al., ICLR 2024 Spotlight, arXiv:2310.17884)	2024	Contextual privacy reasoning	Probing + scenario	GPT-4 leaks 39%, ChatGPT 57%
PrivacyLens (Shao et al., NeurIPS 2024)	2024	Agent privacy norm-following in	Action-based + probing	GPT-4 agent leaks 25.68%

		action		
GAIA (Mialon et al., ICLR 2024, arXiv:2311.12983)	2024	General-AI-assistant capabilities	Exact-match short answers	Humans 92% vs GPT-4-plugins 15% arXiv
AgentBench (Liu et al., ICLR 2024)	2024	Cross-domain agent reasoning	Mixed	Wide capability gap vs proprietary models

The judge-LLM vs. deterministic-state debate. ToolEmu and τ-bench use LLM-emulated environments and LLM judges; AgentDojo and WorkBench use deterministic environment-state checks. AgentDojo argues explicitly that LM-judged evaluation is *itself susceptible to prompt injection* — i.e., the evaluator can be hijacked — and that deterministic environments are necessary for security research. This is a live methodological debate worth highlighting in the seminar. arXiv

What current benchmarks miss:

- Long-horizon (multi-day, multi-session) personalization;
- Cold-start preference acquisition;
- Multi-party coordination with other agents;
- Real-time / interrupt-driven settings (Gaia2, arXiv:2602.11964, begins to address this);
- Cross-cultural / multilingual personal-assistant tasks.

7. Comparison of Leading Research Systems

The seminar's "compare 4–6 systems" axis is best served by the following matrix:

System	Era	Architecture	Memory	Privacy Posture	HITL	Eval
CALO / PTIME (SRI, 2003–2011)	Classic	BDI agent (SPARK) + constraint scheduler + SVM preference learner (PLIANT)	Structured KB	Local desktop; military deployments	PExA "Suggestion Manager" (proactive)	Yearly DAR "admin assistant" te
Generative Agents	Foundational LLM	Memory stream +	Natural-language	N/A (simulation)	N/A	Crowdwork believabilit

(Park et al. UIST 2023)		reflection + planning	stream w/ retrieval			
ReAct (Yao et al. ICLR 2023)	Foundational LLM	Thought–Action–Observation prompt	None (in-context only)	None	None	HotpotQA, ALFWorld, WebShop
MemGPT / Letta (Packer et al. 2023)	Memory-centric	Self-editing OS-style memory	Core / recall / archival tiers	Self-hosted possible	"Heartbeats," interrupts	Multi-sessi chat, doc Q
Mem0 (Chhikara et al. ECAI 2025)	Memory-centric	Extraction + graph-based memory + multi-signal retrieval	Persistent + graph variant (Mem0g)	SOC 2 / HIPAA managed cloud or self-host	API-level	LoCoMo, LongMemE
WorkBench-evaluated ReAct stack (Styles et al. 2024)	Eval target	ReAct + LangChain tools	None	None	None	Outcome-centric, 690 tasks
ScheduleMe (Wijerathne et al. PACLIC 2025)	LLM scheduler	LangGraph supervisor + specialists	Session	OAuth-scoped Google Calendar	None	Zero-shot, 1 tasks
Apple Intelligence + PCC (Apple 2024)	Industry-deployed	On-device ~3B model + PCC offload	App Intents + Personal Context	**Strongest**: stateless cloud, attestation, no-admin servers, verifiable transparency	App Intents permission prompts	Internal
Microsoft 365 Copilot	Industry-deployed	Multi-agent + Graph connectors	Microsoft Graph indexing	Enterprise tenancy, data-residency contracts	Per-action UI confirmation	Internal

Axes worth weighing for the seminar slide:

1. *Single-agent vs. multi-agent* — ReAct vs. ScheduleMe / supervisor patterns.
2. *Memory: ephemeral vs. persistent vs. graph* — ReAct → MemGPT → Mem0.
3. *Privacy posture: prompt-only vs. architectural* — most research systems vs. Apple PCC.
4. *HITL: none vs. explicit confirmation vs. proactive suggestion* — ReAct vs. Apple App Intents vs. PTIME.
5. *Evaluation: LLM-judged vs. deterministic-state* — ToolEmu/τ-bench vs. AgentDojo/WorkBench.
6. *Open-source vs. closed* — Letta/Mem0/AgentDojo vs. Apple/Microsoft.

8. Open Research Challenges

1. **Long-horizon task reliability.** τ-bench's authors report pass^8 strictly below 25% for GPT-4o on retail tasks — that is, even capable agents are non-deterministic on the *same* task across repeated attempts. Reliability — not capability — is the production blocker. arxiv
2. **Multi-tool / multi-domain coordination.** WorkBench's 43% ceiling on GPT-4 across 5 databases illustrates that tool-selection error compounds rapidly with tool count.
3. **Trust calibration and uncertainty communication.** Personal assistants must say "I'm not sure, should I send this?" — research on calibrated agent uncertainty is nascent. Recent work (arXiv:2508.13465) shows agents *know* about risks 95%+ of the time but *fail to act* on that knowledge in execution.
4. **Adversarial robustness.** Prompt injection is unsolved. Defenses degrade utility: AgentDojo reports that tool filtering on GPT-4o "suppresses ASR to 7.5% but often reduces UA (53.3%)" from a baseline of 69%.
5. **Personalization without leaking.** The personalization–privacy tradeoff is unresolved. CI-grounded approaches (Microsoft Research's PrivacyChecker / CI-CoT / CI-RL, 2026) are a promising direction but reduce helpfulness.
6. **Evaluation beyond closed benchmarks.** No current benchmark jointly measures long-horizon personalization, multi-party coordination, adversarial robustness, and privacy norm-following. Gaia2 (2026) is the closest attempt for *dynamic* environments but still does not jointly evaluate privacy.
7. **Temporal reasoning as a missing primitive.** TimeBench, EvolveBench, and TimeShift collectively show that LLMs have a poor grasp of time itself — a foundational deficiency for any calendar agent.

Recommendations

For the seminar deck (~25 slides), the student should structure their pivot from "implementation walkthrough" to "research survey" using the following plan:

Staged narrative arc:

1. *Slides 1–4 — Framing.* Define "personal assistant agent" (Li et al. 2024); contrast Siri-era

classifiers vs ReAct-era tool users.

2. *Slides 5–8 — Architectures.* ReAct → planner-executor → supervisor → memory-augmented, with one diagram per pattern.
3. *Slides 9–12 — Email research.* Smart Reply (2016) → Smart Compose (2019) → LLM email agents (2024–25) → open challenges.
4. *Slides 13–16 — Calendar research.* CALO/PTIME (2011) → x.ai post-mortem (2021) → ScheduleMe (2025) → temporal reasoning failures.
5. *Slides 17–20 — Privacy.* Greshake's indirect prompt injection → AgentDojo/InjecAgent → ConfAIde/PrivacyLens (Contextual Integrity) → Apple PCC as architectural answer.
6. *Slides 21–22 — Personalization.* Generative Agents memory stream → MemGPT → Mem0/A-MEM; the cold-start gap.
7. *Slides 23–24 — Comparison table + benchmark table* (use the two matrices above).
8. *Slide 25 — Open challenges + your take.*

Decisions the student should make now:

- *Pick the memory comparison axis as the deepest dive* (Generative Agents → MemGPT → Mem0). It is the cleanest pedagogical line of research and directly leverages their LangGraph background without becoming an implementation talk.
- *Privacy slide should anchor on three citations*: Greshake et al. AISec 2023, AgentDojo NeurIPS 2024, and PrivacyLens NeurIPS 2024. These are the citations any AI-engineering audience will respect; they collectively tell the "attack → benchmark → norms" story.
- *Use WorkBench's 43% / 3% result as the "reality check" slide* — it is concrete, memorable, and motivates everything that follows.

Benchmarks/thresholds for adjusting emphasis:

- If audience skews security → expand Greshake / AgentDojo / InjecAgent to 4 slides; drop one personalization slide.
- If audience skews HCI / human factors → expand Generative Agents and Park et al.'s evaluation methodology; drop one privacy slide.
- If audience is implementation-curious → add one slide contrasting LangGraph supervisor patterns with AutoGen / Letta, but cap it at one to preserve the research framing.

Caveats

1. **The x.ai story is undocumented in peer-reviewed literature.** Cite it via the 2021 shutdown press coverage and Mortensen's Growth Everywhere / Mixergy interviews; do not represent it as an academic source.
2. **"Control at Stake" (arXiv:2507.02699, 2025) is a preprint** and has not been peer-

reviewed at the time of writing.

3. **The 25–57% leakage numbers from ConfAIde/PrivacyLens depend on benchmark construction** and may overstate real-world leakage; they are best read as relative comparisons across models.
4. **Mem0's reported ~91% latency improvement** comes from the authors' own benchmark (LoCoMo) and against full-context baselines; independent reproduction is limited.
5. **Apple PCC's privacy guarantees rely on Apple's threat model and disclosed binaries**; full source is not public, and independent verification is partial. Some enterprise-compliance commentators note that Apple does not disclose physical PCC node locations, which is a tension with regimes such as GDPR.
6. **Smart Reply's "10% of mobile responses" statistic is from 2016** and reflects the Inbox-by-Gmail era; the comparable 2026 figure is not publicly disclosed.
7. **Generative Agents (Park et al.) is a simulation, not a deployed personal assistant**; cite it for the memory-stream concept, not as direct evidence of personal-assistant performance.
8. **τ-bench is customer-service-focused, not personal-assistant-focused** — extrapolate its findings with care.
9. **CALO/PTIME work pre-dates LLMs**; their preference-learning techniques may not transfer directly but are conceptually instructive for what modern LLM agents *omit*.
10. **AgentDojo's "8% attack success rate under best defenses" figure** is specifically with a secondary attack-detector defense; the same paper notes tool filtering is a different defense that trades utility for security.